Colin Rundel

rundel@gmail.com / colin.rundel@duke.edu

204 Old Chemistry Building • Dept of Statistical Science, Duke University

Research Interests

Applied spatial statistics with a focus on biological and ecological systems, Bayesian statistics, computational methods, statistics and computing education and pedagogy.

Education

University of California, Los Angeles, Department of Statistics

- Ph.D. in Statistics, 2012
- M.S. in Statistics, 2008

Dissertation Topic: Bayesian Methods for Spatial Assignment of Migratory Birds
Advisors: Jan de Leeuw and John Novembre

California Institute of Technology

- B.S. in Biology, 2003

Employment

Associate Professor of the Practice — May 2022 - Present
Department of Statistical Science, Duke University

Assistant Professor of the Practice — June 2015 - May 2022
Department of Statistical Science, Duke University

Lecturer in Statistics and Data Science — May 2019 - May 2021
School of Mathematics, University of Edinburgh

Visiting Assistant Professor / Lecturer — January 2012 - May 2015
Department of Statistical Science, Duke University

Postdoctoral Associate — July 2012 - April 2014
Department of Statistical Science, Duke University

Graduate Student Researcher — September 2010 - December 2011
Novembre Lab, UCLA

Senior Statistical Consultant — March 2009 - December 2011
Statistical Consulting Center, UCLA

Graduate Teaching Assistant — September 2006 - July 2010
Dept. of Ecology and Evolutionary Biology, Dept. of Statistics, UCLA

Teaching

Duke University

- Sta 30 - Statistics and Quantitative Literacy - Fa 12
- Sta 102 - Introductory Biostatistics - Sp 13, Sp 14, Fa 14, Sp 15, Fa 15, Sp 16, Su 16
- Sta 111 - Probability and Statistical Inference - Su 14
- Sta 112 - Better Living through Data Science - Fa 16
- Sta 230 - Probability - Fa 12, Sp 14
- Sta 323 - Statistical Computing - Sp 16, Sp 17, Sp 18, Sp 19, Sp 22, Sp 24, Sp 25, Sp 26
- Sta 344 / 444 / 644 - Spatio-Temporal Modeling - Sp 17, Sp 18, Fa 18, Fa 22, Fa 23
- Sta 523 - Statistical Programming
  Fa 14, Fa 15, Fa 16, Fa 17, Fa 18, Fa 21, Fa 22, Fa 23, Fa 24, Fa 25
- Sta 663 - Statistical Computing and Computation - Sp 22, Sp 23, Sp 25, Sp 26
- Sta 790 - Advanced Statistical Computing - Sp 19

University of Edinburgh
Math 08068 - Facets of Mathematics - Regression Modeling Theme - Fa 19
Math 11176 - Statistical Programming - Fa 19, Fa 20
Math 11205 - Machine Learning in Python - Sp 20, Sp 21

Online Teaching

Coursera - Statistics with R Specialization
Bayesian Statistics
Statistics with R Capstone

Posit Academy
Learn Shiny with R

Publications

Horton, N. J., Alexander, R., Parker, M., Piekut, A., Rundel, C. (2022). *The Growing Importance of Reproducibility and Responsible Workflow in the Data Science and Statistics Curriculum.* Journal of Statistics and Data Science Education, 30(3), 207-208.

Çetinkaya-Rundel, M., Hardin, J., Baumer, B. S., McNamara, A., Horton, N. J., Rundel, C. (2022). *An educator's perspective of the tidyverse.* Technology Innovations in Statistics Education, 14(1).

Poulsen, J., Beirne, C., Rundel, C., Baldino, M., Kim, S., Knorr, J., Minich, T., Jin, L., Núñez, C., Xiao, S., Mbamy, W., Obiang, G., Masseloux, J., Nkoghe, T, Ebanega, M., Clark, C., Fay, M., Morkel, P., Okouyi, J., White, L., Wright, J. (2021). *Long distance seed dispersal by forest elephants.* Frontiers in Ecology And Evolution. 9, 962. DOI.

Beckman M., Çetinkaya-Rundel M., Horton N., Rundel C., Sullivan A., Tackett M. (2020) *Implementing Version Control With Git and GitHub as a Learning Objective in Statistics and Data Science Courses.* Journal of Statistics Education. 29 (Sup 1), 132 - 144. DOI.

Johnson A., Rundel C., Hu J., Ross K., Rossman A. (2020) *Teaching an Undergraduate Course in Bayesian Statistics: A Panel Discussion.* Journal of Statistics Education. 28 (3), 251 - 261. DOI.

Beirne C., Nuñez C., Baldino M., Kim S., Knorr J., Minich T., Jin L., Xiao S., Mbamy W., Obiang G., Masseloux J., Nkoghe T., Ebanega M., Rundel C., Wright J., Poulsen J. (2019) *Estimation of gut passage time of wild, free roaming forest elephants.* Wildlife Biology. 2019 (1).

Cetinkaya-Rundel M., Rundel C.W. (2017) *Infrastructure and tools for teaching computing throughout the statistical curriculum.* The American Statistician. 72 (1), 58 - 65.

Rundel C.W., Schliep E.M., Holland D., Gelfand A. (2015) *A data fusion approach for spatial analysis of speciated* $PM_{2.5}$ *across time.* Environmetrics. 26 (8), 515 - 525.

Rundel C.W., Wunder M., Alvarado A.H., Ruegg K., Harrigan R., Schuh A., Jeffrey K., Siegel R., DeSante D.F., Smith T.B., Novembre J. (2013) *Novel statistical methods for integrating genetic and stable isotope data to infer individual-level migratory connectivity.* Molecular Ecology. 22 (16), 4163 - 4176.

de Bocanegra H.T., Rostovsteva D., Çetinkaya M., Rundel C.W., Lewis C. (2011). *Quality of reproductive health services to limited English proficient patients.* Journal of Health Care for the Poor and Underserved, 22 (4), 1167 - 1178.

Walker D.W., Muffat J, Rundel C.W., Benzer S. (2006). *Overexpression of a Drosophila Homolog of Apolipoprotein D Leads to Increased Stress Resistance and Extended Lifespan.* Current Biology, 16 (7), 674 - 679.

Magazines

Rundel, C.W., Cetinkaya-Rundel M. (2016) La Quinta is Spanish for next to Denny's, Chance 29 (2), 53 - 57

Rundel C.W. (2002) *Genes, Aging, and the Future of Longevity* Engineering & Science, 65 (4), 36 - 40.

Talks & Workshops

Posit Conf 2025 (Workshop) — September 2025
Shiny for R

useR! 2025 — August 2025
Parsing Quarto and R Markdown documents in R

Datascience in the classroom 2025 (Invited) — July 2025
Reflections on a decade of teaching statistical computing

Posit Conf 2024 (Workshop) — August 2024
Introduction to Shiny for R

JSM 2024 (Invited) — August 2024
Programmatic Manipulation of Quarto Documents for Teaching

Posit Conf 2023 (Workshop) — September 2023
Introduction to Shiny

JSM 2023 (Invited) — August 2023
Quarto and automation in teaching statistical computing

Preparing to Teach 2023 — August 2023
Teaching-focused careers

Toronto Workshop on Reproducibility 2022 — 2022
Teaching Statistical Computing with Git and GitHub

ISBA 2022 — 2022
An overview of Bayesian computational frameworks for teaching

RStudio Conf 2022 (Workshop) — July 2022
Introduction to Shiny

ISI Short Course 2021 (Workshop) — June 2021
Teaching Data Science

RStudio Global 2021 — January 2021
parsermd - parsing R Markdown for fun and profit

TLMSCO — September 2020
Teaching computing using git and GitHub

JSM 2020 (Invited) — August 2020
Computation Infrastructure for Teaching Bayesian Modeling

Teaching Statistics and Data Science Online (Online Workshop) — July 2020
Workshop 1: Teaching R online with RStudio Cloud
Workshop 2: Building interactive tutorials in R
Workshop 3: Teaching computing with Git and GitHub

RStudioConf 2020 — January 2020
livecode: broadcast your live coding sessions from and to RStudio

JSM 2019 — August 2019
ghclass: an R package for managing classes with GitHub

JSM 2019 (Workshop) — July 2019
Reproducible Computing

UseR! 2019 — July 2019
ghclass: an R package for managing classes with GitHub

SDSS 2019 (Invited) — May 2019

Using Rocker containers and CI for teaching R-based courses

ICOTS10 2018 (Workshop) — July 2018
Teaching Data Science, Reproducibly

ISBA World Meeting 2018 (Short Course) — June 2018
Reproducible Computing

Joint Statistical Meetings 2017 (Invited) — August 2017
Moving Away from Ad Hoc Statistical Computing Education

UseR! 2017 (Tutorial) — July 2017
Data Carpentry: Open and Reproducible Research with R

Joint Statistical Meetings 2016 (Invited) — August 2016
Statistical Computing as an Introduction to Data Science

UseR! 2016 — July 2016
Continuous Integration and Teaching Statistical Computing with R

Joint Statistical Meetings 2015 — August 2015
Teaching statistical computing leveraging the github ecosystem

UseR! 2015 — July 2015
Teaching R using the github ecosystem

Data Analytics in Business and Social Science Seminar, Duke SSRI — April 2015
Geospatial data and the R ecosystem

Joint Statistical Meetings 2014 — August 2014
A Data Fusion Approach for Space-Time Analysis of Speciated $PM_{2.5}$

Duke Dept of Statistical Science Seminar — February 2014
Using GPUs to improve the computational efficiency of Gaussian process models

Joint Statistical Meetings 2013 — August 2013
GPUs, linear algebra, and efficient computing for Gaussian process models

UseR! 2013 — July 2013
Leveraging GPU libraries for efficient computation of Gaussian process models in R

Joint Statistical Meetings 2012 — August 2012
Leveraging GPU Libraries for Efficient Computation of Bayesian Spatial Assignment Models in R

UseR! 2012 — June 2012
rgeos: spatial geometry predicates and topology operations in R

Joint Statistical Meetings 2011 — August 2011
Spatial Models for Bird Origin Assignment Using Genetic and Isotopic Data

Service

DSS Master's Advisory Committee — Fall 2017 - Spring 2019, Fall 2024 - present

Information Technology Advisory Council — Fall 2017 - present
Chair Fall 2024 - present

DSS Computing Committee — Fall 2014 - present
Chair, Spring 2017 - Spring 2019, Fall 2021 - present

Guest Associate Editor, JSDSE — Fall 2021

Special Issue on Teaching Reproducibility and Responsible Workflow

ASA DataFest @ Duke — Fall 2011 - Spring 2019
Co-organizer

Bayes Impact at Duke — Fall 2014 - Spring 2016

Scientific Registry of Transplant Recipients

Motion Math

Software

`gradermd`: Shiny apps for gradescope like grading of GitHub based assignments

`md4r`: R wrapper of the md4c markdown parsing library.

`checklist`: Tools for automating checking of R projects, with a focus on automated feedback via CI tooling like GitHub actions.

`parsermd`: Tools for parsing and programmatically interacting with R Markdown documents.

`learnrhash`: Tools for recording student results for questions and exercises in learnr documents.

`livecode`: Library for broadcasting source files during live codeing sessions.

`ghclass`: Library for managing classroom and assignment related tasks on github.

`rgeos`: R interface to the Geometry Engine, Open Source (GEOS) library.

`isoscatR`: R package for smoothed and continuous assignment testing (SCAT) of genetic samples

`RcppGP`: Tools for efficiently working with Gaussian Processes in R / C++

`mapnik`: parser and generator for the carto map style language.

Memberships

American Statistical Association